

Individual Interim Report

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Defining the parameter space

To evaluate the estimator, we want to test its performance across a parameter space. Each parameter in should be: interpretable, communicating a particular characteristic of the behaviour of the LGSSM. A condition is defined by a set of parameters: the matrices defining the LGSSM ($\underline{A}$, $\underline{B}$, $\underline{C}$, $\underline{Q}$, and $\underline{R}$); the duration of the simulation, T ; the number of trials, n_{trials} ; the input, u_t ; and the initial state, x_0 .

We can derive additional parameters from these fundamental parameters. From the shapes of the matrices defining the LGSSM we find: the dimension of the latent states, n_x , from the shape of $\underline{A}$, (n_x, n_x) ; the dimension of the inputs, n_u , from the shape of $\underline{B}$, (n_x, n_u) ; and the dimension of the observations, n_y , from the shape of $\underline{C}$, (n_y, n_x) . These dimensions form new scalar parameters that we can vary.

We can also derive the spectral radius of $\underline{A}$, $\rho(\underline{A})$, which communicates the stability, stationarity, and memory of the LGSSM.

We can derive the controllability and observability Gramians, $\underline{W}_c$ and $\underline{W}_o$ respectively. We can use the determinant, trace, minimum eigenvalue, and the condition number.

We can derive the input-to-state gain

We can derive the observation gain

We can derive the observation ratio

We can directly vary the signal types and initial conditions.

Analytically generating a set of conditions that vary themselves uniformly across a high dimensional parameter space of these derived quantities would require a large amount of code; instead we propose a strategy to randomise the generated conditions, and keep randomly generating new conditions until the parameter space is effectively traversed. These conditions can then be saved to disk and reused